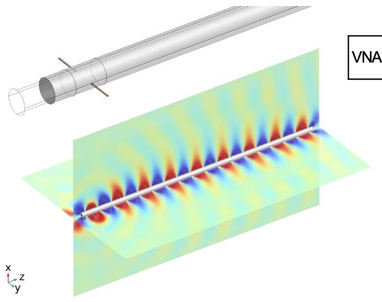
UPISC UNIVERSITY OF PITTSBURGH INFRASTRUCTURE SENSING COLLABORATION WORKSHOP

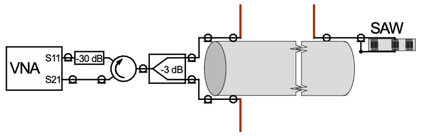
NATIONAL ENERGY TECHNOLOGY LABORATORY

Electromagnetic Wave on Pipeline structure for Nondestructive Evaluation (NDE) application

Wenbo Li [1], David Greve [2], Paul Ohodnicki [1]
[1] University of Pittsburgh
[2] Carnegie Mellon University

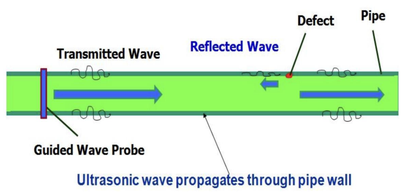
INTRODUCTION



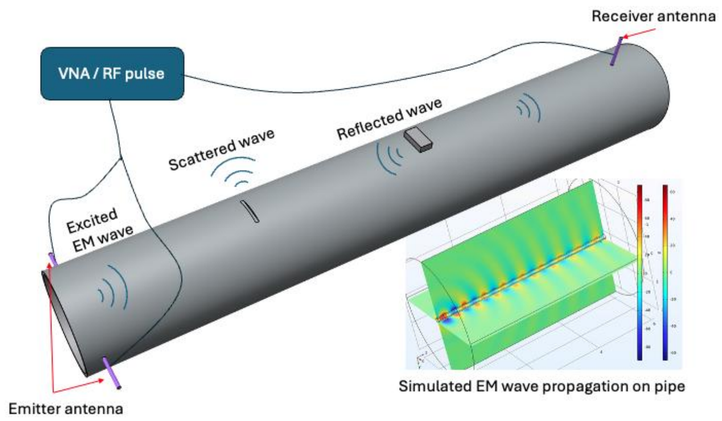


- Goubau wave travels on surface of metal pipe
- Goubau wave used for interrogation of SAW devices at certain frequency

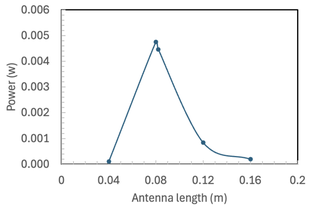
	Ultrasound	EM wave
Target feature	Metal perturbation	Surroundings & Metal
Frequency	~100 kHz	~5 GHz
Launching / sensing	Piezo / Fiber	Cone launchers/ Antennas



METHODOLOGIES

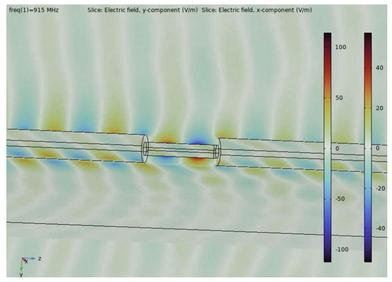


Simulation results



- Antenna length influence on energy level

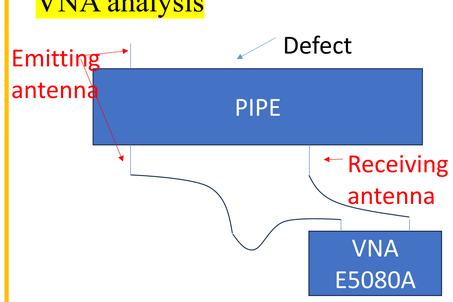
40% material removed:

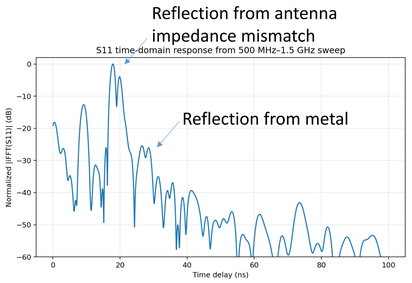


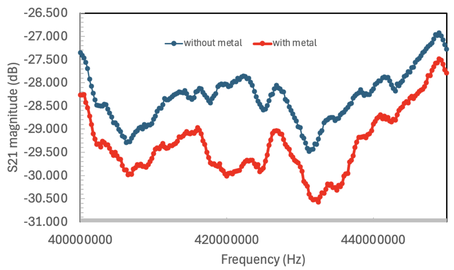
36.44% change in S21 power

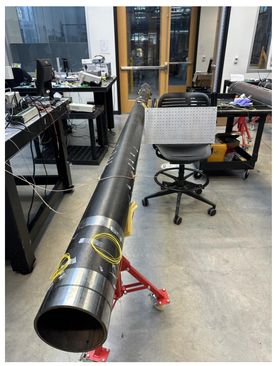
PRELIMINARY RESULTS & PERSPECTIVES

VNA analysis



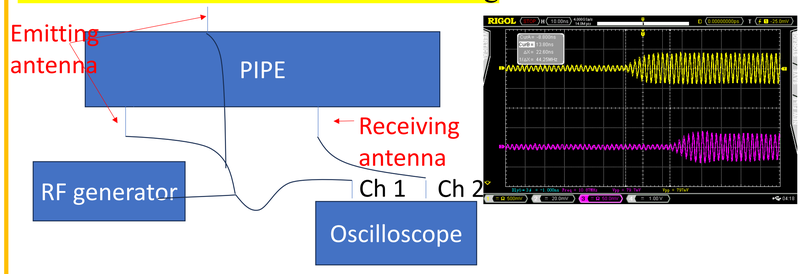






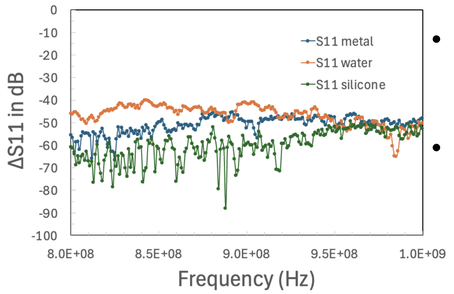
- Attenuation tells feature size
- iFFT of frequency tells location

RF modulation for time domain reading



- RF pulse delay on receiver matches distance on pipe, confirming Goubau wave existence
- Received signal with reflector has longer ramp up time, confirming reflection from reflector antenna

Defect detection



- Change in attenuation signal with 3 different material on pipe
- Water patch gives highest response



Acknowledgement

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